

STATEMENT OF WORK

Intent to Sole Source: Solid-State Power Controller

1.0 Introduction/Background: The System-Agnostic Arc Fault Implementor (SAFDI) testbench at the Glenn Research Center (GRC) is being used to design DC arc fault detectors for eventual use on hybrid-electric aircraft systems, on spacecraft power systems, and in microgrids. SAFDI is part of the Subsonic Vehicle Technologies and Tools (RVLT) project. The SVTT project is exploring multiple technologies that would enable electric or hybrid-electric aircraft, resulting in quieter and more fuel-efficient flight as well as opening new air transportation markets.

SAFDI is expanding its capability to include a high-voltage switch to interrupt arcs quicker than any COTS electromechanical switch at planned higher test voltages and currents, which is not possible with available equipment. This procurement is for one (1) solid state power controller, which will enhance the lab's ability to demonstrate the viability of the system-agnostic DC series arc fault detector design process across multiple power system topologies.

2.0 Scope of Work: This procurement is for (1) solid-state power controller.

Specifications include:

- Continuous voltage rating:
 - o Greater than or equal to 1kVDC
- Current ratings:
 - o Greater than or equal to 200A continuous forward current @ 85°C
 - o 600A interrupting current
- Inductive load capability
 - o Greater than or equal to 150 μ H @ 600 A
- Galvanic isolation:
 - o Greater than or equal to 2500Vrms
- On-state resistance ($R_{DS,ON}$)
 - o Less than or equal to 1.5m Ω @ 25°C
- Temperature range
 - o -40°C to +85°C
- Discrete indicators:
 - o Load status
 - o Trip status
 - o Overtemperature status
- Discrete controls:
 - o On/Off

- Trip Reset
- Mass:
 - No more than 1 lb.
- Dimensions:
 - No larger than 7.00" x 4" x 1.50"
- Trip features:
 - Adjustable overcurrent protection (I^2T)
 - Short circuit protection
- Connections
 - DB15

3.0 Objectives: Enhance the existing capability of the SAFDI testbench to arc at higher powers and with different load impedances. This addition will lend credibility to the system-agnostic design process claim.

4.0 Description of the Work/Contractor Tasks: The contractor task consists of providing the equipment. No meeting or reviews are expected for this off-the-shelf equipment.

5.0 Deliverables: The deliverable for this procurement is the (1) solid-state power controller.